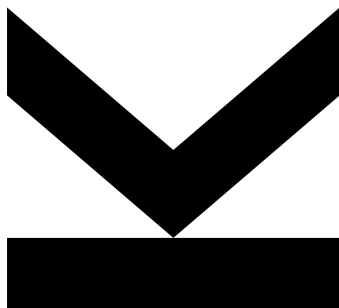


Reproduction and Analysis of Meta-Pec for Personalized Vehicle Energy Consumption Estimation



Practical Work in Artificial Intelligence
Personalized Vehicle Energy Consumption Estimation
Machine Learning Institute

Abstract This report reproduces and analyzes the Meta-Pec framework for personalized vehicle energy consumption estimation. The work studies the original method, runs experiments on the available datasets, and compares Meta-Pec with two simpler baselines: Multiple Linear Regression and XGBoost. In my reproduced setup, XGBoost achieved the strongest results, while Meta-Pec was harder to train and did not fully reproduce the performance reported in the original paper.

Keywords Meta-Pec, vehicle energy consumption estimation, personalized prediction, meta-learning, XGBoost

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Abstract

Accurate vehicle energy consumption estimation is important for efficient trip planning, sustainable transportation, and intelligent mobility systems. A major challenge in this task is that energy usage is influenced not only by route and road conditions, but also by the behavior of individual drivers. This makes personalized estimation an important but difficult problem.

This report studies the paper “A Preference-aware Meta-optimization Framework for Personalized Vehicle Energy Consumption Estimation” by Lai et al. (2023), which proposes the Meta-Pec model for personalized vehicle energy consumption prediction. The framework combines driving preference learning from historical trajectories, selection-based driving behavior prediction, vehicle energy consumption estimation, and a meta-learning strategy for driver-specific adaptation.

As part of this practical work, I studied the paper, analyzed the proposed method, and ran my own experiments on the provided datasets and code-base. In addition to Meta-Pec, I implemented two classical baselines, namely Multiple Linear Regression (MLR) and XGBoost, to compare a complex personalized deep learning method with simpler tabular approaches. My experiments showed that XGBoost achieved the strongest results in my setup on both datasets, while Meta-Pec was considerably more difficult to train and did not reproduce the performance reported in the original paper under the reduced training budget.

Story Summary Q&A

What is the central question?

How can vehicle energy consumption be predicted more accurately when driver-specific behavior is taken into account? The paper studies whether learning latent driving preferences from historical trajectories and combining them with meta-learning can improve personalized vehicle energy consumption estimation.

Why is this question important?

Vehicle energy consumption estimation is relevant for route planning, battery-aware decision making, and sustainable transportation. Since different drivers can consume very different amounts of energy even on similar routes, personalization is important for realistic prediction.

What evidence or data are needed to answer this question?

The task requires historical trips, trajectories, route information, vehicle-related information, and ground-truth trip energy consumption. The paper evaluates the method on two real-world datasets, namely VED and ETTD.

What methods are used to get this evidence or data?

The paper uses two real datasets, preprocesses the trajectories into trips, extracts route and feature representations, and then applies the Meta-Pec framework for prediction. In my own work, I additionally implemented MLR and XGBoost baselines on the same split structure used in the available codebase.

What analyses are applied to the data?

The paper evaluates models using MSE, MAE, and MAPE, and also performs ablation studies, long-tail analysis, parameter sensitivity analysis, and a case study. In my own experiments, I compared the implemented baselines and Meta-Pec on both datasets using the same evaluation metrics.

What evidence supports the answer?

In the original paper, Meta-Pec achieves the best reported performance among the compared methods on both datasets. In my own experiments, however, XGBoost performed best, while Meta-Pec was harder to reproduce reliably and showed instability across users.

What is the answer?

The paper argues that personalized preference learning and meta-optimization improve vehicle energy consumption estimation. My own experiments suggest that the idea is interesting and well motivated, but the reproduced performance depends strongly on the implementation and training setup.

How does this change what we know?

The paper broadens the task from general energy estimation to personalized energy estimation and shows a meta-learning-based way to address long-tail users. My practical work adds the insight that strong baselines such as XGBoost can still outperform a complex model when reproduction conditions are limited.

1. Introduction

Vehicle Energy Consumption (VEC) estimation aims to predict how much energy a vehicle will need for a trip before the trip starts. This task is relevant for route planning, battery-aware navigation, cost reduction, and sustainable transportation. A reliable estimate can help both drivers and mobility systems make better decisions.

The difficulty is that energy consumption depends on several interacting factors, such as route length, road properties, speed, acceleration, traffic conditions, departure time, and vehicle type. In addition, different drivers may consume different amounts of energy even under similar conditions because of differences in driving style. This makes personalized estimation important.

The paper studied in this report addresses this problem through Meta-Pec, a preference-aware meta-optimization framework for personalized vehicle energy consumption estimation. The main idea is to learn latent driver preferences from historical trajectories and combine them with route information. The model also uses a meta-learning strategy to adapt to drivers with limited historical data.

The aim of this practical work was to evaluate how well the main ideas of Meta-Pec can be understood and reproduced in a constrained practical setting. More specifically, I wanted to understand the model architecture, run the available Meta-Pec implementation, and compare it with two simpler baselines: Multiple Linear Regression and XGBoost. This comparison was useful because it tested whether a complex personalized deep learning method provides a practical advantage over strong tabular methods in my setup.

The report first introduces the background and problem definition. It then summarizes the Meta-Pec framework, describes the experimental setup, and presents the results from both the original paper and my own experiments. Finally, the report discusses the limitations, reproducibility challenges, and main lessons from the practical work.

2. Background and Problem Definition

2.1 Vehicle Energy Consumption Estimation

Vehicle Energy Consumption estimation predicts the amount of energy a vehicle will require for a future trip. It is useful for route planning, energy-efficient driving assistance, battery management, and sustainable mobility. The task is difficult because energy consumption is not controlled by one variable, but by a combination of route, vehicle, environmental, and behavioral factors.

Traditional approaches often rely on numerical or physics-based models. These can be interpretable, but they usually depend on simplified assumptions and handcrafted variables. Machine learning methods are more flexible because they can learn nonlinear patterns directly from data. However, many existing methods focus mainly on general route or trip features and do not fully model how individual drivers differ.

2.2 Personalized Estimation

Meta-Pec treats VEC estimation as a personalized prediction problem. The motivation is simple: two drivers may travel on the same route but consume different amounts of energy because of different acceleration, braking, speed, and lane-changing behavior. Therefore, a good model should not only learn a global mapping from route features to energy consumption, but should also consider the driver’s historical behavior.

A further challenge is the long-tail setting. Some drivers have many recorded trips, while others have only a few. A robust personalized model should use rich historical information when it is available and still adapt when driver-specific data is limited. This is one reason why Meta-Pec uses meta-learning.

2.3 Problem Definition

In the setting of the paper, a route is represented as an ordered sequence of road segments. A trajectory describes how a vehicle moves along a route and contains temporal, spatial, and vehicle-state information. Each trip has a total energy consumption value, which is the prediction target.

For a driver u , the model receives a target route R_c and historical trips H_u . The goal is to predict the total energy consumption y of the future trip:

$$f_{\theta} : (R_c, H_u) \rightarrow y,$$

where θ denotes the model parameters. This formulation highlights the central challenge: the prediction must combine route-specific information with personalized information extracted from the driver’s past behavior.

3. Related Work

3.1 Numerical Methods

Early VEC estimation methods often used numerical or physics-based models. These approaches describe energy usage through factors such as speed, acceleration, road grade, distance, and traffic conditions. Their main advantage is interpretability, but they are limited when many factors interact in nonlinear ways. They also often treat drivers only coarsely, for example by using driver groups instead of learned driver-specific representations.

3.2 Data-driven Methods

Data-driven methods learn from observed trip and trajectory data. Classical models such as linear regression, decision trees, support vector machines, and gradient boosting use aggregated features like average speed, distance, or acceleration. Deep learning models, such as LSTMs, GRUs, encoder-decoder models, and Transformers, can process sequential route or trajectory data more directly. Their weakness is that personalization is often handled through simple driver statistics rather than through learned latent driving preferences.

3.3 Meta-learning and Personalization

Meta-learning is useful when many related tasks exist but each task has limited data. In this paper, each driver can be viewed as a separate task. The goal is to learn shared knowledge that can quickly adapt to a specific driver. This idea is related to model-agnostic meta-learning introduced by Finn et al. (2017). Meta-Pec uses this motivation to improve personalized prediction, especially for long-tail drivers with limited history.

4. The Meta-Pec Framework

4.1 Framework Overview

Meta-Pec is a preference-aware meta-optimization framework for personalized VEC estimation. It combines route information with personalized driver information extracted from historical trips. Instead of treating all drivers equally, the model tries to capture latent driving preferences and use them for prediction.

The framework has four main components. First, a driving preference learning module extracts driver preferences from historical trajectories. Second, a selection-based driving behavior prediction module estimates likely driver-specific behavior on the target route. Third, a VEC estimation module combines route features, predicted behavior, and preference information to predict total energy consumption. Finally, a driver-specific meta-optimization module improves adaptation across drivers.

Figure 1 shows the overall structure of the Meta-Pec framework.

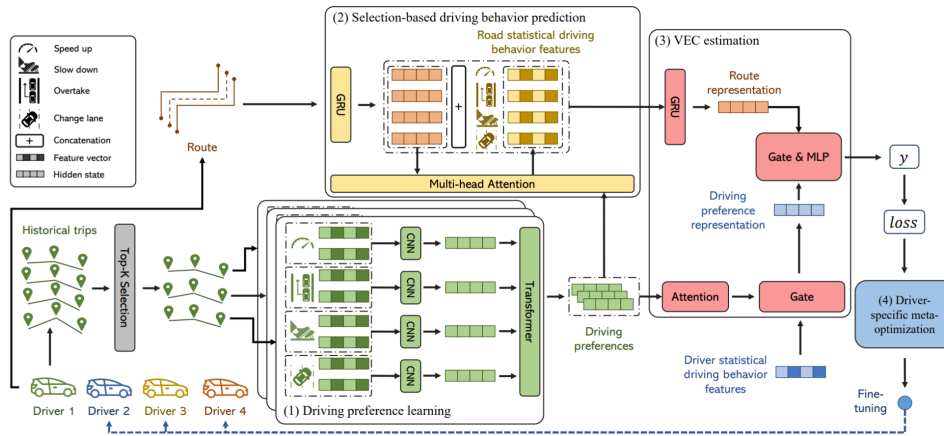


Figure 1: Overall architecture of the Meta-Pec framework. The model combines historical driver trips, driving preference learning, selection-based behavior prediction, VEC estimation, and driver-specific meta-optimization. Source: Lai et al. (2023).

4.2 Feature Construction

Meta-Pec uses several feature groups to represent trip context, driver behavior, vehicle state, road properties, and vehicle type. Before entering the model, these features are projected into a shared embedding space. The main feature groups are summarized in Table 1.

Table 1: Main feature groups used in the Meta-Pec framework.

Feature group	Examples	Role in the model
Trip features	Month, departure time, route length	Provide global context about the target trip before the route sequence is processed.
Statistical driving behavior features	Average speed, acceleration, energy consumption per hour, energy consumption per kilometer	Represent macroscopic driver behavior and are also used as supervision signals in behavior prediction.
Vehicle state features	Local time, speed, acceleration, energy consumption per hour, energy consumption per kilometer	Describe fine-grained trajectory states and help learn latent driving preferences.
Road features	Road type, one-way indicator, number of lanes, maximum speed, road length	Represent the structure and properties of the target route.
Vehicle-type features	Vehicle category or vehicle type embedding	Allow the final prediction to account for differences between vehicle types.

4.3 Driving Preference Learning

A key part of Meta-Pec is learning driver preferences from historical trajectories. The model first augments vehicle-state information with distance-time encodings, using sine and cosine functions to preserve spatiotemporal structure. The trajectory is then split into smaller segments, and a CNN extracts local behavior patterns from each segment. These representations are passed through a Transformer encoder to obtain a unified driver preference representation.

This design allows the model to learn more than simple average statistics. It can represent local behavior patterns such as acceleration changes and speed trends, while still combining them into one driver-level preference representation.

4.4 Selection-based Driving Behavior Prediction

After learning driver preferences, Meta-Pec predicts likely behavior on the target route. Instead of using all historical trips equally, it selects the top-K most

relevant trips based on route similarity and temporal similarity. This helps the model focus on past trips that are more likely to be useful for the current prediction.

The target route is encoded with a GRU, and multi-head attention combines the route representation with the selected historical preference representations. The output is a prediction of road-level statistical driving behavior features. This connects historical driver behavior with the specific route of the current trip.

4.5 VEC Estimation

The final VEC estimation module combines the predicted road-level behavior, the route representation, statistical driving behavior features, and the learned driver preference representation. A gating mechanism fuses these sources of information, and an MLP produces the final energy consumption estimate. The model also uses a vehicle-type embedding so that differences between vehicle types can influence the final prediction.

4.6 Driver-specific Meta-optimization

The meta-optimization component addresses the problem that some drivers have little historical data. In the paper, each driver is treated as a separate task. During meta-training, the model updates on a support set and is evaluated on a query set. The query losses across drivers are used to improve the shared initialization. After meta-training, the model is fine-tuned for each driver. The goal is to learn parameters that can adapt quickly to individual drivers.

5. Experimental Setup

5.1 Datasets

The paper evaluates Meta-Pec on two real-world datasets: the Vehicle Energy Dataset (VED) and the Electric Taxi Trajectory Dataset (ETTD). VED contains 25,661 trips from 348 drivers, while ETTD contains 18,546 trips from 377 drivers. The reported average route lengths are 4.81 km for VED and 5.10 km for ETTD. A compact comparison is shown in Figure 2.

Category	VED	ETTD
# of trips	25,661	18,546
# of drivers	348	377
Time span	11/1/2017-11/7/2018	10/22/2014
# of total distance	130,864 km	87,251 km
Avg. route length	4.81 km	5.10 km
Avg. speed	38.59 km/h	13.54 km/h

Figure 2: Dataset statistics for VED and ETTD, including number of trips, number of drivers, time span, total distance, average route length, and average speed. Source: Lai et al. (2023).

VED was collected in Ann Arbor, Michigan, using OBD-II loggers and includes vehicle trajectories, speed, and energy-related signals. ETTD contains electric taxi trajectories from Shenzhen, China. In ETTD, energy consumption is not measured directly from battery logs but is calculated as required mechanical energy at the wheel. The two datasets differ in geography, vehicle types, and recording conditions, which makes them useful for evaluating generalization.

5.2 Preprocessing

The original paper preprocesses the raw trajectories by splitting them into trips whenever a driver stops for more than five minutes. The route of each trip is extracted through map matching. Energy consumption is computed at GPS sample level and then aggregated to obtain the trip-level target value. For ETTD, energy consumption must be estimated indirectly because the dataset mainly provides speed and coordinate information. The authors also anonymize drivers through masked identifiers.

5.3 Evaluation Metrics

The paper evaluates performance using Mean Squared Error (MSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). MSE penalizes large errors strongly, MAE gives an interpretable average absolute error, and MAPE expresses error relative to the true value:

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|.$$

MAPE can exceed 100% when the prediction error is larger than the true value. It can also become unstable when ground-truth values are very small. This is important when interpreting the high MAPE value of MLR on VED.

5.4 Implementation Details

The paper reports an embedding size of 20 and $\text{top-K} = 5$ selected historical trips. The CNN filter size is $q = 4$, and both the Transformer encoder and multi-head attention use 4 heads. The GRU and MLP hidden sizes are 20 and 40. Adam is used for optimization, with different inner-loop, outer-loop, and fine-tuning learning rates. For each driver, 10% of the records are used for validation, 20% for testing, and the rest for training.

In my own experiments, I used the available Meta-Pec codebase and implemented MLR and XGBoost baselines. The experiments were run on a local Windows laptop with an Intel Core i7-10510U CPU, 16 GB RAM, and integrated Intel UHD Graphics. No dedicated GPU was used. For Meta-Pec, I used a reduced training setup with 6 epochs instead of the full training schedule reported in the paper. The reduced Meta-Pec run took approximately 30 minutes. Console output and experiment files were used to record the run, and the final test metrics were used for this report.

For the baselines, I used the same general split structure as the provided project code, but converted the raw samples into fixed-size tabular feature vectors. MLR and XGBoost were much faster and easier to train than Meta-Pec. Therefore, my experiments should be interpreted as a practical reproduction under limited compute, not as a full-scale reproduction of the original training setup.

5.5 Code and Reproducibility Documentation

To support reproducibility, the submitted project files include the code used for the practical experiments, including the Meta-Pec run and the two baseline implementations. The baseline scripts convert the samples into tabular feature vectors and evaluate MLR and XGBoost using the same general split structure as the project code. The values reported in Table 3 are taken from the final test outputs.

Exact reproduction of Meta-Pec depends on the available codebase, processed dataset files, random seeds, and training configuration. Since the model contains several interacting modules and user-specific fine-tuning steps, I report the results as a practical reproduction under limited compute rather than as a complete independent reimplementations.

5.6 Baselines

The original paper compares Meta-Pec with several numerical and data-driven baselines, including Average, MLR, XGBoost, DNN, LSTM, Transformer, LD-FeRR, Enc-Dec, PLd-FeRR, and Meta-TTE. In my own work, I focused on MLR and XGBoost because they were feasible to implement within the available time and code structure. MLR serves as a simple linear baseline, while XGBoost is a strong nonlinear model for structured tabular data.

Table 2: Comparison of the models used in my experiments.

Model	Type		Purpose in this report
MLR	Linear baseline	regression	Tests how much can be achieved with a simple linear model and aggregated features.
XGBoost	Nonlinear model	tabular	Provides a strong classical machine learning baseline for structured features.
Meta-Pec	Personalized learning model	deep	Represents the main method proposed in the original paper and tests the personalized meta-learning approach.

6. Results

6.1 Results Reported in the Original Paper

In the original paper, Meta-Pec is compared with ten baseline methods on VED and ETDD. The reported results show Meta-Pec as the strongest model in terms of MSE, MAE, and MAPE. The authors use these results, together with ablation, long-tail, parameter sensitivity, and case study experiments, to argue that preference learning and meta-optimization improve personalized VEC estimation.

6.2 Results from My Own Experiments

Table 3 summarizes the results from my own experiments. In my setup, XGBoost achieved the strongest performance on both datasets. On VED, it clearly outperformed MLR and Meta-Pec. On ETDD, the same pattern appeared again, with XGBoost giving the lowest error values among the models I ran.

Meta-Pec did run successfully, but my reproduced results did not match the superiority reported in the original paper. On VED, Meta-Pec performed better than MLR in MAE and MAPE, but worse than XGBoost. On ETDD, Meta-Pec performed worse than both baselines. The weaker Meta-Pec result should be interpreted carefully because my run used only 6 epochs and took approximately 30 minutes on a CPU-based laptop without dedicated GPU acceleration.

Another observation from the logs is that Meta-Pec behaved unevenly across users. Some users had small prediction errors, while others produced large errors that strongly affected the final average. This instability was especially visible during user-specific fine-tuning and was one of the clearest practical differences between Meta-Pec and the simpler baselines.

Table 3: Average per-user test results from my own experiments. For Meta-Pec, the reported MAPE values were converted from fractional form to percentages for consistency with the baseline implementations.

Dataset	Model	MSE	MAE	MAPE (%)
VED	MLR	0.0668	0.1043	123.49
VED	XGBoost	0.0199	0.0208	12.66
VED	Meta-Pec	0.0727	0.0607	32.86
ETTD	MLR	0.3870	0.3996	34.58
ETTD	XGBoost	0.2203	0.2905	23.48
ETTD	Meta-Pec	1.1183	0.6744	55.13

The MAPE value of 123.49% for MLR on VED is possible because MAPE is a relative error measure. If the absolute error is larger than the true value, the percentage error can exceed 100%. Also, very small ground-truth energy values can make MAPE unstable. For this reason, MAPE should be interpreted together with MSE and MAE rather than alone.

6.3 Ablation Study

The original paper includes an ablation study that removes or changes parts of Meta-Pec, such as meta-optimization, personalization modules, behavior prediction, and historical trip selection. The full model performs best in the paper, which supports the claim that the modules contribute to the final performance. I did not reproduce the full ablation study, but it remains important for understanding why the model is built in a modular way.

6.4 Long-tail Driver Analysis

A main motivation of Meta-Pec is the long-tail setting, where some drivers have only a small number of historical trips. The paper reports that driver-specific meta-optimization improves performance for such drivers. The importance of personalization is also visible in the variation of average energy consumption across drivers, shown in Figure 3.

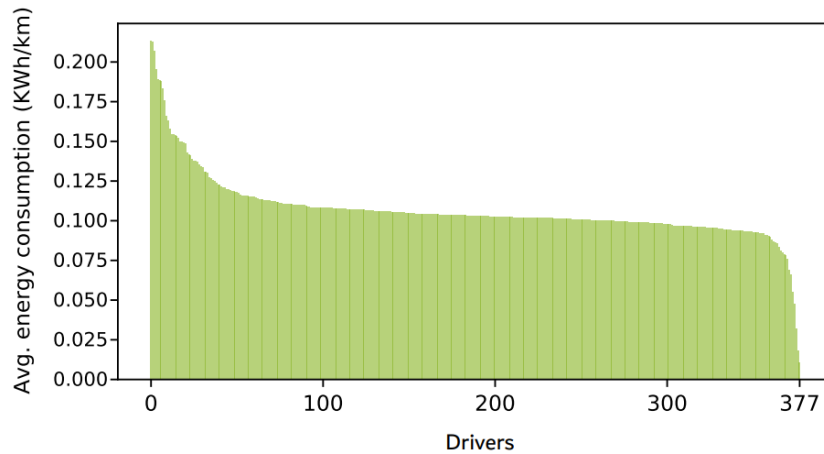


Figure 3: Average energy consumption per driver, sorted by consumption level. The variation across drivers supports the motivation for personalized vehicle energy consumption estimation. Source: Lai et al. (2023).

My own experiments also showed user-level variation. Some users were predicted relatively well, while others produced much worse errors. This does not fully reproduce the paper’s long-tail experiment, but it supports the general point that user adaptation is central to this task.

6.5 Parameter Sensitivity

The paper studies sensitivity to hyperparameters such as the number of selected historical trips K and behavior-segment size q . I did not run a full sensitivity analysis, but the training behavior already showed that Meta-Pec is sensitive to training budget, user-level fine-tuning, and implementation details. This supports the practical observation that the model is powerful but not easy to tune.

6.6 Case Study

The original paper also includes a qualitative case study showing ground-truth and predicted energy consumption along a route. I did not reproduce this visualization, but it is useful because it shows that the model is intended not only to predict one trip-level value, but also to capture meaningful route-dependent behavior patterns.

7. Critical Assessment

The strongest part of the paper is its motivation. The authors correctly identify that vehicle energy consumption is not only a route-level or traffic-level problem, but also a personalized behavior problem. This makes the paper conceptually strong and gives the method a clear research identity.

Another strength is the structure of the model. Meta-Pec does not add personalization as a small extra feature. Instead, it builds the method around driver preference learning, selective use of similar historical trips, and adaptation across users. The long-tail perspective is also valuable because real systems often contain many users with limited data.

The main weakness is practical reproducibility. The framework combines preference learning, behavior prediction, route modeling, gating, and meta-optimization. This creates many points where training can become unstable. My experiments made this visible because XGBoost was much easier to train and performed better in my limited setup.

A second limitation is my reduced experimental budget. The Meta-Pec run used only 6 epochs and was executed on a CPU-based laptop without a dedicated GPU. Therefore, the reproduced Meta-Pec results should not be interpreted as a definitive evaluation of the method’s full potential. They should be interpreted as a practical reproduction under limited compute.

Overall, I would say that the paper is strong in motivation, originality, and method design, but more challenging from a reproduction perspective. The gap between the clean paper narrative and the effort required to obtain strong results is an important limitation to acknowledge.

8. Discussion

The main lesson from this work is that complex research models and practical reproducibility are not the same thing. Meta-Pec is a meaningful and ambitious framework. Its core idea is convincing because driving style clearly matters for energy usage. The attention to long-tail drivers is also relevant, since real-world systems often contain users with very uneven data availability.

At the same time, my experiments show that simpler baselines should not be underestimated. XGBoost performed best on both datasets in my setup. This does not prove that the original paper is wrong, but it shows that under constrained reproduction conditions, a simpler model can be more stable and effective than a complex deep learning architecture.

The user-specific adaptation step is both one of the most interesting parts of Meta-Pec and one of the most fragile parts during reproduction. Some users had good results, while others caused large errors. This suggests that personalized fine-tuning can be powerful, but it also increases sensitivity to data splits, training settings, and implementation details.

Therefore, the reproduced results should be interpreted honestly. My work supports the motivation of personalized VEC estimation and shows that user-level variation is a real issue. However, it also shows that Meta-Pec requires careful tuning and enough compute to be fairly evaluated. In a limited practical setting, strong tabular baselines may be more reliable.

9. Conclusion

In this report, I studied Meta-Pec from both a theoretical and practical perspective. The paper proposes a personalized VEC estimation model that combines driving preference learning, selection-based behavior prediction, VEC estimation, and driver-specific meta-optimization. Its main idea is to move beyond coarse handcrafted personalization and learn richer driver-dependent representations from historical trajectories.

My own experiments produced a more mixed picture than the original paper. I successfully ran MLR, XGBoost, and Meta-Pec on the available datasets. In my setup, XGBoost achieved the strongest performance on both VED and ETDD. Meta-Pec did not reproduce the superiority reported in the paper. This result should be interpreted carefully because my Meta-Pec run used only 6 epochs, took approximately 30 minutes, and was performed on a local CPU-based laptop without dedicated GPU acceleration.

Overall, the practical work was valuable because it showed both the promise and the difficulty of modern machine learning research. Meta-Pec is creative and well motivated, but reproducing such a model is significantly harder than understanding it on paper. The main lesson is the importance of careful reproduction, strong baselines, transparent limitations, and honest evaluation in applied machine learning.

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